

Network Analysis

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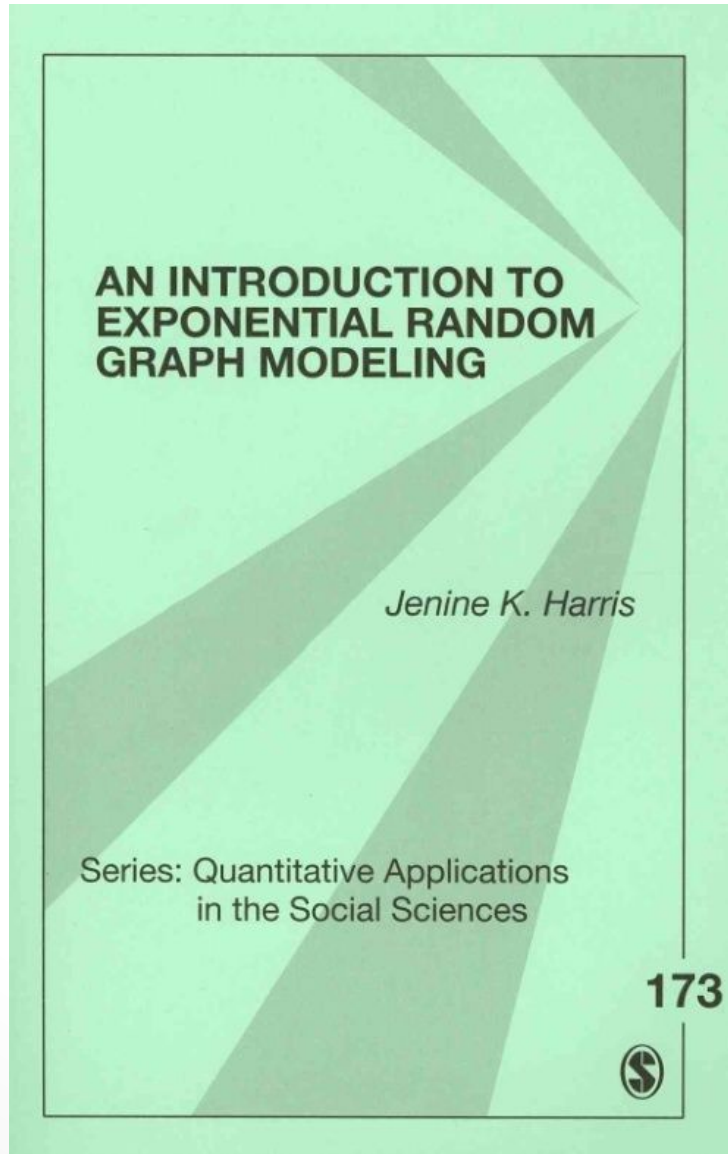
Spring 2017

Exponential Random Graph Models (ERGMs)

...

It's like logistic regression for networks...

(Highly) Recommended Reference
<100 pages. Easy Read.



ERGMs

Used to understand whether a given observed network can be derived from characteristics of network members and the pattern of ties comprising the whole network.

Timeline for Development

1959

- Erdős & Rényi: Simple Random Graph
- Nodes are connected randomly with probability = density.

1981

- Holland & Leinhardt: p_1 model
- Assumes dyadic independence (Flawed)

1986

- Frank & Strauss: Markov Random Graph Model
- Allows for adjacent edges to be statistically dependent.

1996

- Wasserman & Pattinson: p^* model
- Capable of introducing covariates

2006

- Pattinson & Robbins, Snijders, Hancock & Hunter et al.
- Social circuit dependence by adding terms to p^* model

p^* : logit models for social networks

What influences are driving the formation of links in a social network?

- Self-organization and Social/structural balance
 - Reciprocity
 - Transitivity
- External factors
 - Node level of attributes
 - Gender
 - Age
 - Dyad level attributes measuring homophily
 - Same gender
 - Number of projects shared in the past

p^* : logit models for social networks

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➤ External factors

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To model these effects, we'll use information about number of mutual ties, number of triangles, etc.

p^* : logit models for social networks

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➤ External factors

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To model these effects, we'll use node-level and edge-level variables

Independence?

What independence

- Problem: No MLE
- Solution 1: Pseudo-likelihood estimation (convenient)
- Better Solution: Markov Chain Monte Carlo (MCMC)
MLE (principled)

The p^* model

$$P(Y_{ij} \mid n \text{ actors}, Y_{ij}^c) = \text{logistic}(\vartheta_1 \delta_{z_1(y)} + \vartheta_2 \delta_{z_2(y)} + \dots)$$

Model Parameters.
Interpreted like
logistic parameters.

Change Statistics for each
edge, each variable. This is
the data underlying the
model.

Probability of a tie
between node i and
node j , conditional on
the rest of the
network held
constant.

The p^* model

- If each edge is a case to be predicted in a manner similar to logistic regression, then our input data will look as follows:

Edge (i,j)	Y_{ij}	$\delta - z_1(y_{ij})$	$\delta - z_2(y_{ij})$	$\delta - z_3(y_{ij})$
(1,2)	1	5	4	3
(1,3)	1	5	4	9
(1,4)	0	3	2	0
(1,5)	0	2	0	6
(1,6)	1	11	1	0
...	...	...	...	...

Change Statistics $\delta_{z(y_{ij})}$

- $\delta_{z(y_{ij})}$ represents the change in the network statistic $z(y)$ when an edge is added between node i and node j .
- This is the key difference in the interpretation of an ERGM and a classic logistic regression.

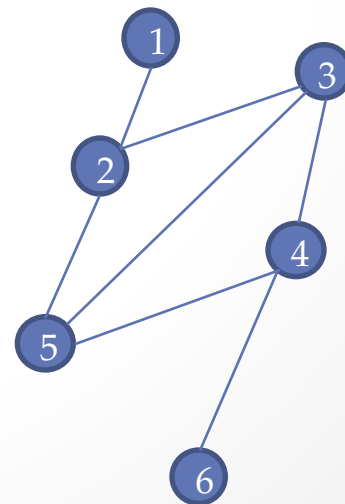
Change Statistics

- The change statistic answers the following question:
- How will the network statistic in question change if the (i,j) edge is included in the network vs. not included.

Change Statistics for Network Statistics

- Network statistics such as **number of triangles** are included to help a model learn self-organizational features like transitivity.
- For $z_1(y) = \text{number of triangles}$, $\delta_{z_1}(y_{ij})$ measures the increase in the number of triangles if edge (i,j) is present vs. absent and all else in network held constant.

Edge	$\delta_{z_1}(y_{ij})$
(1,2)	0
(1,3)	1
(3,5)	2
(3,4)	1



Change Statistics for Edge-level Variables

- Suppose you have edge attribute like number of joint projects in past.
- Change statistic $\delta_{z_2}(y_{ij})$ is simply the value of the edge covariate.
- Measures how the TOTAL number of past joint projects in the network changes if this edge is included

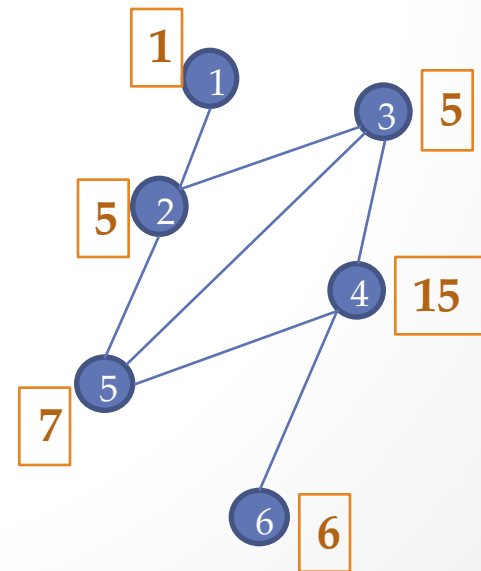
Number of Past Projects for each Pair						
	1	2	3	4	5	6
1	0	1	2	0	1	0
2	1	0	3	0	1	0
3	2	3	0	1	1	2
4	0	0	1	0	0	0
5	1	1	1	0	0	1
6	0	0	2	0	1	0

Edge	$\delta_{z_2}(y_{ij})$
(1,2)	1
(1,3)	2
(3,5)	1
(3,4)	1

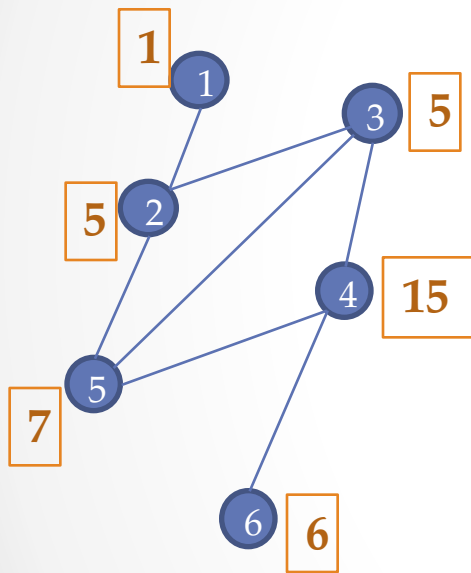
Change Statistics for Node-level Variables

- Suppose you have node-level variable like tenure.
- Change statistic $\delta_{z_3}(y_{ij})$ is the sum of the tenure for the two nodes.
- Measures how the TOTAL tenure of the network changes if this edge is included.

Edge	$\delta_{z_3}(y_{ij})$
(1,2)	6
(1,3)	6
(3,5)	12
(3,4)	20



Summary: Input Data



Edge	Y_{ij}	$\delta_{z_1(y_{ij})}$ (Transitivity)	$\delta_{z_2(y_{ij})}$ (Past Proj)	$\delta_{z_3(y_{ij})}$ (Tenure)
(1,2)	1	0	1	6
(1,3)	0	1	2	6
(3,5)	1	2	1	12
(3,4)	1	1	1	20
...	...	...	...	...

One row in this table for all possible pairs (i, j)

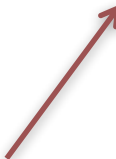
Change in
number of
triangles in the
network when
edge is added

Edge variable
for Past
Projects

Sum of
tenures of the
two adjacent
nodes

Change Statistics (Data Underlying Model)

Edge	δ - Tenure	δ -Past Projects	δ -ESP
(1,3)	5	4	3
(2,6)	5	4	9
(3,2)	3	2	0
(3,4)	2	0	6
(3,9)	11	1	0



Sum of
tenures of the
two adjacent
nodes



Edge variable
for Past
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Change in
number of
ESPs in the
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Homophily Terms (Like interactions)

- **Homophily**: "love of the same." Tendency of individuals to associate with similar others.
- Homophily change statistic for categorical variable:
 - $\delta_{ij} = 1$ if node i and node j have the same value for cat. variable
 - $\delta_{ij} = 0$ if they do not.
 - This is equivalent to an edge variable that says whether the two nodes match characteristics.
- These are the “nodematch” terms in ergm package

Model Output

```
=====
Summary of model fit
=====
```

```
Formula:   FInet ~ edges + nodecov("NumLeads")
```

```
Iterations: 5 out of 20
```

```
Monte Carlo MLE Results:
```

	Estimate	Std. Error	MCMC %	p-value
edges	-3.54963	0.11645	0	<1e-04 ***
nodecov.NumLeads	0.23146	0.01813	0	<1e-04 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Null Deviance: 6890 on 4970 degrees of freedom
Residual Deviance: 2704 on 4968 degrees of freedom
```

```
AIC: 2708    BIC: 2721    (Smaller is better.)
```

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Not your average Maximum
Likelihood Estimation. And not as
fast ☹️

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Intercept. Baseline probability of an edge. ($\exp(\beta_0) = \text{density}$)

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Node Covariate. As *NumLeads* increases, so do odds of an edge.

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These p-values are simulated. What is probability this variable appears significant if no true relationship?



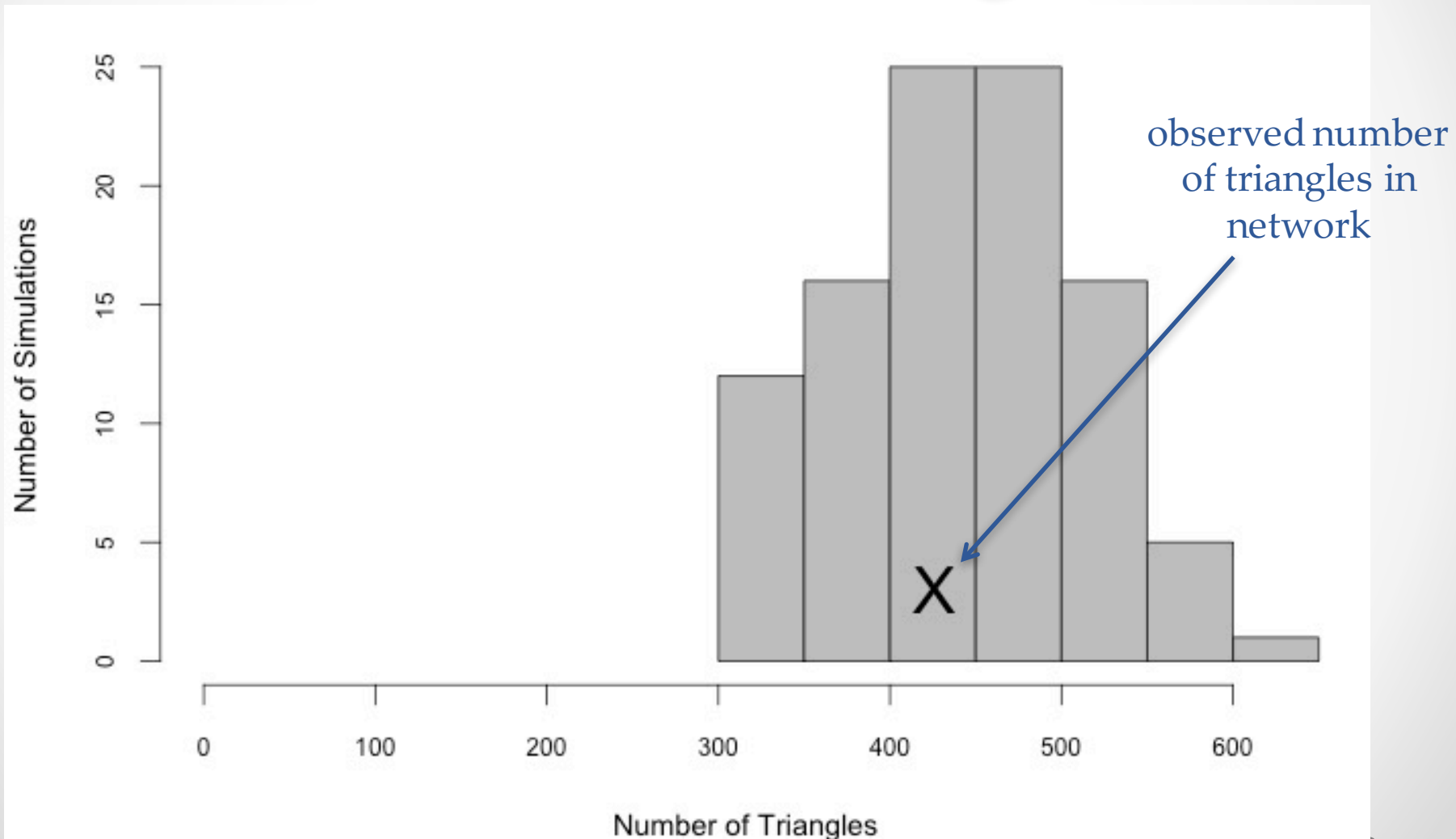
Model Evaluation (Comparing Models)

- -2Log-Likelihood (“Deviance”)
 - Smaller is better
 - Will always get smaller when parameters added to model
 - Can only be used to compare nested models
- AIC + BIC
 - Variations on Deviance that penalize models for having additional parameters, p .
 - $AIC = Deviance + 2p$
 - $BIC = Deviance + p \cdot \ln(N)$
 - Smaller is better
 - Can be used to compare non-nested models.

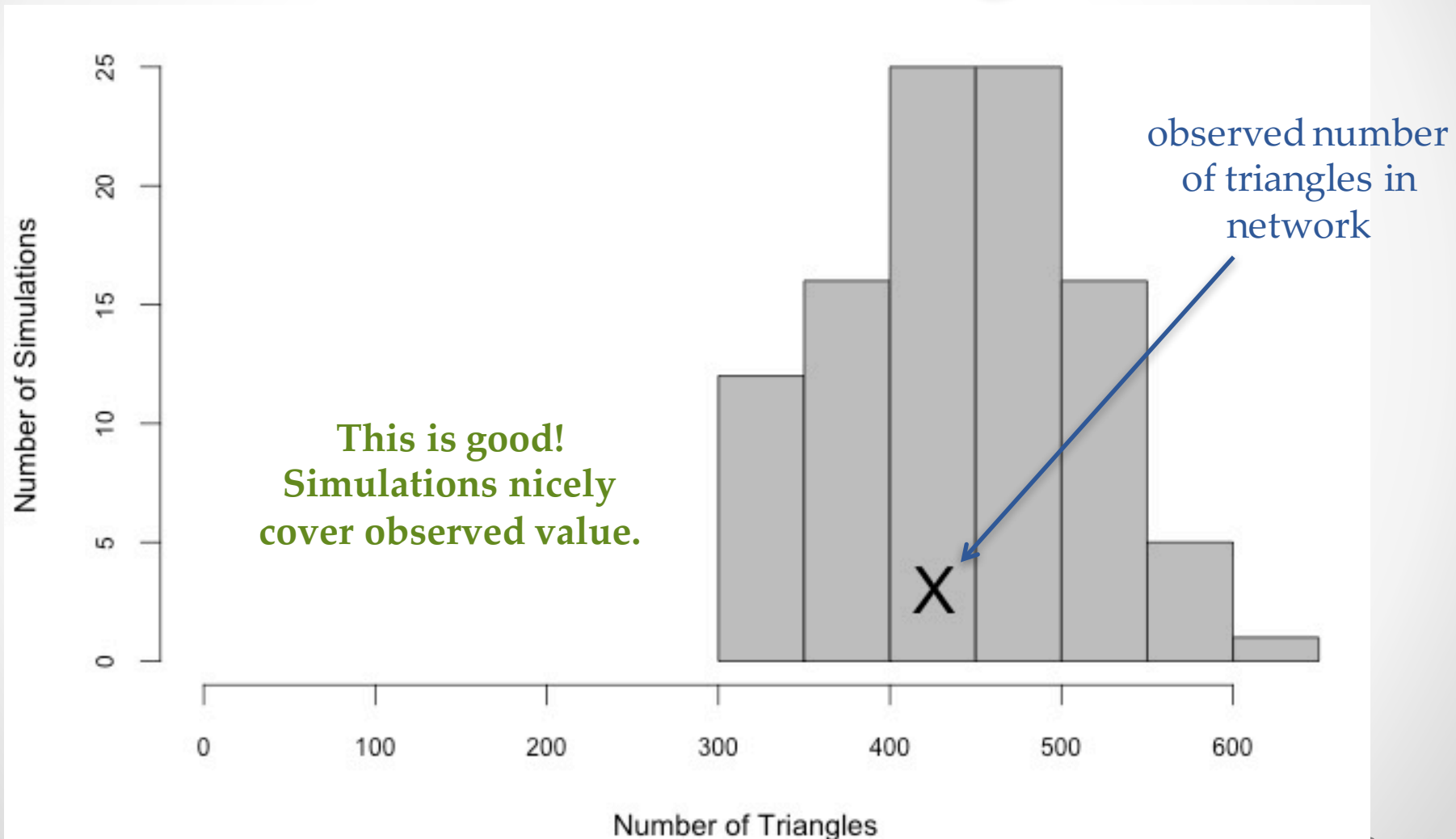
Model Evaluation (Assessing Network Fit)

- Want to assess how well your model captures the observed network's attributes
- If we simulate networks using our model, will they...
 - Match the degree distribution of our network?
 - Have a comparable number of triangles? (i.e. similar transitivity)
 - Have a comparable distribution of geodesic distances?
- Particularly if we're interested in *prediction*, these assessments will be key.

Example: Assessing Network Fit (Number of Triangles)



Example: Assessing Network Fit (Number of Triangles)



Example: Assessing Network Fit (Degree Distribution)

<u>Degree</u>	<u>Observed</u>	<u>Min Sim</u>	<u>Mean Sim</u>	<u>Max Sim</u>	<u>p-value</u>
0	6	1	7.5	9	0.55
1	27	9	11.8	26	0
2	17	8	15.1	21	0.15
3	6	5	8.2	12	0.04
4	3	0	0.03	3	0.03
5	2	0	0.02	2	0.0001

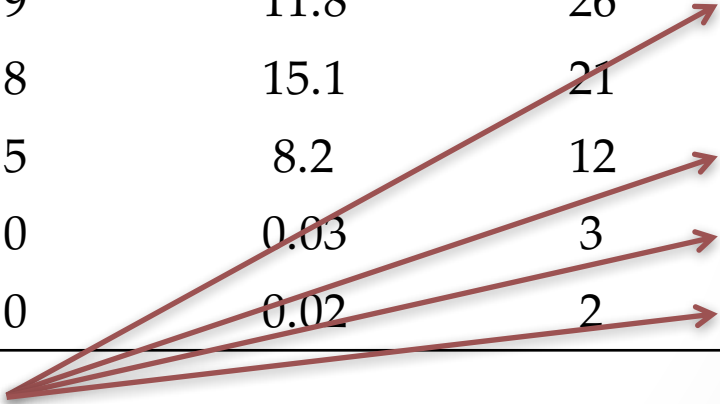
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Good. It is very likely we would simulate a graph with the observed number of degree 0 and 2 nodes using our model.

Example: Assessing Network Fit (Degree Distribution)

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3	6	5	8.2	12	0.04
4	3	0	0.03	3	0.03
5	2	0	0.02	2	0.0001



Bad. It is very unlikely we would simulate a graph with the observed number of degree 1,3,4,5 nodes using our model.

(Although we can't expect perfection here. Want to do *pretty* well for degrees with large sample sizes)

Fortune Interactive Consulting

- 71 Consultants
- Most projects are brought in by the consultants themselves (some by sales/support staff)
- Each project has 1 or 2 leads, managing a team of 3 or 6 individuals respectively.
- Each lead is responsible for creating a team of 3 other consultants to handle the project.
- Encouraged to work with many different consultants, not always draft same team
 - (They had been worried that newly hired consultants were not being utilized.)
 - Some accusations of preferential treatment based on mutual interests

Fortune Interactive Consulting

- Projects can range in duration from 1 month to 24 months
- Consultants have at most 4 projects ongoing at any given time.
- Question: Can we determine factors that contribute to this network structure? Are mutual interests in sports driving professional relationships?